

Prompt Sensitivity and Its Effect on the Stability of LLM-Generated Research Conclusions

Abstract

Large Language Models (LLMs) have rapidly become useful tools for literature synthesis, hypothesis generation, data interpretation, scientific writing, and research assistance. Their usefulness, however, depends not only on model capability but also on how a research task is formulated in the prompt. Prompt sensitivity refers to changes in model behavior caused by variations in prompt wording, structure, examples, ordering, or contextual information, even when the underlying task remains semantically equivalent. This phenomenon creates a significant methodological concern for scientific applications: if small prompt changes produce materially different interpretations, evidence summaries, or conclusions, then an LLM-generated conclusion may not be sufficiently stable to support scientific inference.

This paper reviews the emerging literature on prompt sensitivity, consistency, reasoning, hallucination, and evaluation of large language models, with particular emphasis on implications for research conclusions. Existing studies demonstrate that LLM outputs can vary substantially under paraphrased or template-modified prompts, although the magnitude of this effect depends on the model, task, evaluation method, and prompt construction. Recent work has introduced quantitative measures such as POSIX and PromptSensiScore and has proposed sensitivity and consistency metrics for systematic evaluation. At the same time, research suggests that some reported prompt sensitivity may be an artifact of rigid evaluation procedures rather than genuine semantic instability. Techniques including chain-of-thought prompting, self-consistency, few-shot demonstrations, structured prompts, retrieval augmentation, and multi-sample aggregation can reduce some forms of variability, but they do not guarantee factual or inferential reliability.

The paper argues that conclusion stability should be treated as a distinct evaluation dimension from accuracy. A research conclusion can be factually plausible yet unstable across semantically equivalent prompts. A robust scientific workflow should therefore evaluate prompt perturbation, semantic agreement, evidential consistency, factual grounding, uncertainty, and reproducibility together. The paper identifies major research gaps, including the absence of standardized benchmarks specifically designed for scientific conclusions, insufficient separation of prompt-induced variability from sampling variability, limited longitudinal studies across model updates, and inadequate measurement of inferential stability. Future research should develop prompt-invariant scientific evaluation protocols, conclusion-level stability metrics, uncertainty-aware aggregation methods, and human-in-the-loop verification frameworks.

Keywords

Large Language Models; prompt sensitivity; prompt robustness; research conclusions; scientific reasoning; LLM reliability; reproducibility; hallucination; prompt engineering; consistency evaluation.

1. Introduction

Large Language Models (LLMs) have evolved from general-purpose language generation systems into widely used research assistants capable of summarizing literature, extracting information, generating hypotheses, writing code, interpreting textual evidence, and assisting with scientific communication. The emergence of increasingly capable foundation models has expanded the range of tasks that can be performed through natural-language interaction rather than task-specific programming or fine-tuning (Bommasani et al., 2021).

The increasing use of LLMs in research introduces an important methodological question: Can an LLM be expected to generate the same research conclusion when the same research question is expressed using slightly different prompts? In conventional scientific research, equivalent formulations of a hypothesis should not

fundamentally alter the empirical conclusion if the underlying evidence and analytical procedure remain unchanged. For LLM-based workflows, however, the wording and structure of the prompt are themselves part of the computational input. Consequently, apparently superficial changes can influence the generated response.

Prompt sensitivity has been formally investigated in recent research. Chatterjee et al. (2024) introduced POSIX, a Prompt Sensitivity Index designed to quantify changes in model behavior under intent-preserving prompt variations. Zhuo et al. (2024) proposed ProSA and the PromptSensiScore metric to investigate instance-level prompt sensitivity and its relationship with model confidence. Errica et al. (2025) further distinguished sensitivity, concerning changes in predictions under prompt rephrasings, from consistency, concerning whether examples belonging to the same class receive stable predictions.

The issue becomes more consequential when LLMs are used to produce research conclusions rather than simple classifications. A classification error may affect one label, whereas instability in a research conclusion can alter the interpretation of an entire body of evidence. For example, prompts asking whether a treatment is effective, whether an observed relationship supports a hypothesis, or whether the literature demonstrates a particular causal mechanism may produce different levels of confidence or even different substantive conclusions under small wording changes.

This concern is connected to broader limitations of LLMs. Language models can produce fluent but factually unsupported statements, commonly described as hallucinations (Ji et al., 2023). TruthfulQA demonstrated that even highly capable language models can reproduce plausible human misconceptions rather than consistently produce truthful answers (Lin et al., 2022). Therefore, consistency alone cannot establish correctness. Conversely, an apparently correct answer generated by one prompt does not establish that the conclusion is robust.

This paper examines the relationship between prompt sensitivity and the stability of LLM-generated research conclusions. It addresses five central questions: (1) What is prompt sensitivity, and what factors produce it? (2) How does prompt sensitivity differ from ordinary generation stochasticity and factual error? (3) How can prompt sensitivity influence research conclusions? (4) Which current methods can reduce or detect conclusion instability? and (5) What methodological framework is required for reliable LLM-assisted scientific research? The central argument is that research conclusion stability should become a first-class evaluation criterion for LLM-based scientific workflows, alongside accuracy, factuality, relevance, and reasoning quality.

2. Background and Related Work

2.1 Large Language Models and Prompt-Based Task Specification

Modern LLMs are generally trained using large-scale language modeling objectives and subsequently adapted for instruction following and interaction. GPT-3 demonstrated that scaling language models can produce strong few-shot performance without task-specific parameter updates, with the task being specified primarily through textual prompts and demonstrations (Brown et al., 2020).

This property makes prompts an unusually important component of the computational pipeline. In a traditional machine-learning system, a trained classifier receives a structured input according to a predefined interface. In an LLM system, the interface itself can contain natural-language instructions, examples, formatting requirements, role descriptions, contextual documents, constraints, and questions. Changing any of these components can modify the model's probability distribution over possible outputs.

Consequently, prompt engineering has become a major area of research. Chain-of-thought prompting, for example, demonstrated that explicitly requesting intermediate reasoning steps can substantially improve performance on several reasoning benchmarks (Wei et al., 2022). Zero-shot chain-of-thought further demonstrated that a short instruction asking the model to reason step by step can substantially change performance (Kojima et al., 2022).

2.2 Prompt Sensitivity

Prompt sensitivity can be defined as the degree to which an LLM's output changes in response to a meaning-preserving modification of its prompt.

Let a research task be represented by $P=(I,Q,C,F)$, where I represents instructions, Q the research question, C the contextual evidence, and F the requested output format. Suppose P' is a semantically equivalent perturbation of P . Ideally, semantic equivalence between P and P' should imply semantic equivalence between the corresponding outputs. In practice, this implication does not always hold.

Chatterjee et al. (2024) showed that even intent-preserving changes such as wording, spelling, and template modifications can produce measurable changes in model likelihoods and outputs. Zhuo et al. (2024) similarly found that prompt sensitivity varies across tasks and models and reported evidence that larger models may exhibit greater robustness under some conditions. Recent theoretical work by Liu and Chu (2026) models prompt sensitivity through changes in the model's probability distribution and argues that meaning-preserving prompts can nevertheless occupy substantially different representations.

2.3 Reasoning and Research Conclusions

The relevance of prompt sensitivity becomes greater when prompts ask models to perform multi-step reasoning. Chain-of-thought prompting has shown that the explicit elicitation of reasoning can improve performance on mathematical, symbolic, and commonsense tasks (Wei et al., 2022; Kojima et al., 2022). However, a performance improvement on a benchmark does not necessarily imply that the generated reasoning is scientifically valid.

Research conclusions usually involve several stages: Evidence → Interpretation → Reasoning → Conclusion. Prompt variation can potentially affect each stage, including which evidence the model emphasizes, how evidence is categorized, the level of uncertainty expressed, which causal explanation is preferred, whether contradictory evidence is discussed, and the final strength of the conclusion.

3. Prompt Sensitivity and Stability of Research Conclusions

3.1 From Output Variability to Conclusion Variability

Not every difference between two LLM responses represents a meaningful instability. Two responses may use different wording while expressing the same conclusion. Therefore, lexical similarity is insufficient for evaluating scientific stability.

For scientific use, a useful distinction is:

Lexical Stability $\neq$ Semantic Stability $\neq$ Inferential Stability.

Lexical stability measures similarity of wording. Semantic stability measures whether outputs communicate equivalent meanings. Inferential stability measures whether the conclusions, evidence interpretation, confidence, and implications remain substantively equivalent. The third level is most important for research.

3.2 Sources of Prompt-Induced Instability

Surface form: Minor lexical changes can influence token probabilities and model behavior. POSIX specifically identifies spelling, wording, and prompt-template variations as sources of sensitivity (Chatterjee et al., 2024).

Prompt structure: The ordering of instructions, evidence, examples, and questions can affect information use. Research on long contexts has shown that LLM performance can deteriorate when relevant information is positioned in the middle of a long context rather than near the beginning or end (Liu et al., 2024).

Demonstrations and examples: Few-shot examples influence how a model interprets the intended task. Zhuo et al. (2024) report that few-shot examples can reduce prompt sensitivity in some settings, although demonstrations can also introduce unintended biases.

Reasoning instructions: Instructions such as “explain step by step,” “critically evaluate the evidence,” or “provide a concise answer” can change the reasoning trajectory. Chain-of-thought and zero-shot reasoning research demonstrates that seemingly simple prompting changes can produce substantial performance differences.

Decoding stochasticity: Prompt sensitivity should not be confused with generation randomness. Even with the same prompt, probabilistic decoding can generate different responses. Observed variability can therefore be conceptually decomposed into prompt, sampling, model, context, and evaluation components.

4. Measuring Stability of LLM-Generated Research Conclusions

4.1 Prompt Perturbation Testing

A straightforward method is to construct a family of semantically equivalent prompts $P=\{P_1,P_2,\dots,P_n\}$, where all prompts express the same research task. The model produces corresponding outputs, which can then be compared using semantic similarity and conclusion-level criteria.

A conceptual prompt-stability score can be defined as $S_p = 1 - (1/N) \sum D(C_i, \bar{C})$, where C_i represents the conclusion extracted from output i , $\bar{C}$ is the aggregate conclusion, and D is a normalized semantic or inferential distance.

4.2 Sensitivity Versus Consistency

Errica et al. (2025) distinguish sensitivity, which captures how much predictions change under prompt rephrasing, from consistency, which examines whether equivalent examples receive stable predictions. For research conclusions, both dimensions are relevant.

A model may have high sensitivity and low consistency; low sensitivity and high consistency; low sensitivity but incorrect consistency; or high sensitivity while still producing a reliable aggregate. Therefore, stability cannot replace factual validation.

4.3 Conclusion-Level Evaluation

Research workflows should evaluate factuality, relevance, semantic stability, inferential stability, and uncertainty calibration. Accuracy and stability are orthogonal: a model may be correct but unstable, stable but incorrect, both correct and stable, or neither.

5. Effect on Scientific Research Conclusions

5.1 Literature Reviews

LLMs are increasingly used to summarize large collections of academic papers. Prompt sensitivity can influence which themes are emphasized and which limitations are reported. Prompts emphasizing “major findings” may encourage positive summaries, whereas prompts emphasizing “limitations and contradictory evidence” may generate more cautious conclusions. If these differences are caused by framing rather than by the underlying evidence, the resulting literature review may be systematically biased.

5.2 Hypothesis Evaluation

Consider a researcher asking whether the literature supports a hypothesis H . A positively framed prompt could encourage evidence supporting H , while a skeptical prompt could encourage attention to contradictory studies. A reliable research assistant should distinguish evidence for H from prompt framing.

5.3 Causal Interpretation

Causal reasoning is especially vulnerable because research papers often distinguish correlation from causation. A prompt that explicitly asks for causal mechanisms may induce the model to construct a stronger causal

narrative than the evidence warrants. LLMs should therefore distinguish association, prediction, intervention, causal mechanism, temporal precedence, and causal identification.

5.4 Research Recommendations

Prompt sensitivity can also affect recommendations for future work. An LLM may propose different research gaps depending on whether the prompt emphasizes novelty, feasibility, societal impact, theoretical contribution, or methodological limitations. Si, Yang, and Hashimoto (2024), in a large study involving more than 100 NLP researchers, found that LLM-generated ideas could be judged highly novel while also identifying limitations in feasibility, diversity, and self-evaluation.

6. Current Approaches and Developments

6.1 Chain-of-Thought Prompting

Chain-of-thought prompting encourages explicit intermediate reasoning and has demonstrated gains on several reasoning benchmarks (Wei et al., 2022). For scientific research, structured reasoning can help separate evidence extraction, interpretation, limitations, synthesis, and conclusion. However, detailed reasoning does not guarantee correct reasoning.

6.2 Self-Consistency

Self-consistency samples multiple reasoning paths and selects an answer supported by the most consistent reasoning paths (Wang et al., 2023). Conceptually, this changes a single output into an aggregate over multiple reasoning trajectories. It can reduce sensitivity to individual decoding trajectories, but shared systematic bias can remain.

6.3 Prompt Ensembles

Researchers can construct an ensemble of semantically equivalent prompts and derive a final conclusion from agreement across prompts. Outputs can be categorized as supports, partially supports, insufficient evidence, contradicts, or uncertain.

6.4 Few-Shot Prompting

Few-shot examples can constrain task interpretation. Zhuo et al. (2024) report that demonstrations may alleviate prompt sensitivity in certain settings. For research applications, examples should be balanced to avoid confirmation bias.

6.5 Retrieval-Augmented Generation

Retrieval-augmented approaches can ground responses in explicitly supplied sources. They are useful for research because the model can distinguish retrieved evidence from prior knowledge. However, retrieval does not eliminate prompt sensitivity because the model may still interpret the same evidence differently.

6.6 Structured Evidence Tables

Forcing the model to extract evidence before producing a conclusion can reduce narrative drift. A structured table can include study, sample, finding, limitation, and support for the hypothesis.

6.7 LLM-as-a-Judge

Zheng et al. (2023) showed that strong LLM judges can correlate substantially with human preferences in open-ended evaluation while also identifying biases such as position and verbosity. For scientific conclusions, relying on one LLM to judge another creates a risk of correlated errors.

6.8 Stability-Aware Prompt Optimization

Recent work treats prompt stability as a quality criterion. Chen et al. (2026) proposed semantic stability for evaluating auto-generated prompts and argued that prompt quality should account for response consistency rather than immediate task performance alone. This suggests a shift from optimizing prompts for accuracy alone toward accuracy, stability, and reliability.

7. Research Challenges and Limitations

7.1 Lack of a Standard Definition of Stability

Different studies use different notions of consistency, sensitivity, robustness, semantic similarity, and reproducibility. A standardized taxonomy should distinguish prompt sensitivity, decoding variability, contextual sensitivity, factual instability, inferential instability, and evaluation instability.

7.2 Semantic Equivalence Is Difficult to Establish

Two prompts can be linguistically different but semantically equivalent, while two superficially similar prompts can contain subtle differences in scope. Benchmark construction therefore requires human-validated semantic equivalence rather than automatic paraphrasing alone.

7.3 Evaluation Metrics May Create Artificial Sensitivity

Hua et al. (2025) provide an important counterpoint: some apparent prompt sensitivity may result from evaluation procedures themselves. Rigid answer matching and likelihood-based evaluation can exaggerate differences when responses are semantically equivalent.

7.4 Correctness Does Not Follow from Consistency

A model that repeatedly generates the same false conclusion is highly consistent but scientifically unreliable. TruthfulQA demonstrates that LLMs can reproduce false beliefs and misconceptions despite fluent responses (Lin et al., 2022).

7.5 Model Updates and Version Drift

Commercial and open models can change through updates, fine-tuning, safety adjustments, or inference changes. Scientific reproducibility therefore requires recording model name and version, date of inference, system prompt, user prompt, retrieved evidence, temperature and decoding settings, number of samples, output, and evaluation method.

7.6 Domain-Specific Risks

Prompt sensitivity may have different consequences across medicine, social science, engineering, and policy research. Generic prompt-stability benchmarks may therefore not adequately represent scientific risk.

8. Research Gaps and Future Directions

8.1 Scientific Conclusion Stability Benchmarks

There is a need for benchmarks specifically targeting research conclusions. A scientific benchmark could provide peer-reviewed papers, a research question, semantically equivalent prompts, expert-validated evidence summaries, expected conclusion categories, uncertainty labels, and expert judgments of acceptable inference.

8.2 A Conclusion Stability Index

Future work could develop a standardized Conclusion Stability Index (CSI) combining semantic agreement, evidential agreement, uncertainty agreement, and factual consistency. Such a metric would distinguish similar language from genuinely stable scientific inference.

8.3 Stability Under Evidence Perturbation

Prompt robustness should be studied together with evidence robustness by independently varying prompt, evidence, and sampling factors. This would help identify whether instability originates from wording, evidence selection, reasoning, decoding, or interaction effects.

8.4 Calibration of Scientific Uncertainty

Future LLM systems should be evaluated not only on whether the final conclusion is correct, but also on whether confidence changes appropriately when evidence changes. Strong evidence should lead to higher confidence, mixed evidence to moderate confidence, and limited evidence to lower confidence.

8.5 Adversarial Prompt Perturbation

Researchers should deliberately test synonyms, sentence restructuring, instruction order, positive versus neutral framing, negative versus neutral framing, role prompts, output formats, number of examples, and evidence-order changes.

8.6 Human-in-the-Loop Verification

Scientific use should remain human supervised for consequential conclusions. A robust workflow can require the LLM to provide a conclusion, supporting evidence, contradictory evidence, uncertainty, assumptions, limitations, and source references.

8.7 Provenance and Reproducibility

Every AI-assisted research result should ideally have an auditable provenance record: Research Question → Prompt → Evidence → Model → Output → Evaluation.

8.8 Beyond Prompt Engineering

Future scientific systems may combine LLMs with symbolic reasoning, statistical inference, retrieval systems, citation verification, formal hypothesis testing, structured databases, uncertainty estimation, and human expert review. The objective should be to move from prompt-dependent generation toward evidence-grounded scientific inference.

9. Conclusion

Prompt sensitivity represents a significant but underappreciated challenge for the use of LLMs in scientific research. Existing research demonstrates that relatively small changes in prompt wording, templates, demonstrations, and contextual organization can alter model behavior. At the same time, recent work cautions that some apparent sensitivity may arise from imperfect evaluation methodologies rather than genuine semantic instability.

For ordinary conversational applications, such variability may be relatively harmless. In scientific research, however, it can become consequential because LLMs are increasingly used to summarize evidence, evaluate hypotheses, identify research gaps, generate interpretations, and formulate conclusions. If semantically equivalent prompts yield materially different conclusions from the same evidence, then the resulting output cannot be considered fully robust.

The key distinction is between output similarity and conclusion stability. Scientific reliability requires more than consistent wording. It requires that factual claims, evidence interpretation, uncertainty, reasoning, and final inference remain reasonably stable under meaning-preserving perturbations.

Current approaches—including chain-of-thought prompting, self-consistency, few-shot examples, prompt ensembles, retrieval augmentation, structured evidence extraction, and stability-aware prompt optimization—provide useful mechanisms for reducing or diagnosing variability. However, none independently guarantees scientific correctness. Hallucination, evaluation artifacts, confirmation bias, model updates, and correlated errors remain important limitations.

Future research should establish standardized scientific benchmarks for prompt robustness, develop conclusion-level stability metrics, separate prompt sensitivity from sampling and context effects, and integrate factual verification with uncertainty calibration. The most reliable paradigm is unlikely to be one in which an LLM is simply given a carefully engineered prompt and trusted to produce a conclusion. Instead, LLM-assisted science should be designed as a reproducible, evidence-grounded, multi-stage process in which prompt robustness is explicitly measured and conclusions are independently verified.

Ultimately, prompt sensitivity should not be regarded merely as a prompt-engineering inconvenience. It is a methodological property of LLM-based scientific workflows. Treating it as such is essential if LLMs are to contribute to research without weakening the principles of reproducibility, transparency, evidential reasoning, and scientific reliability.

References

1. Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., et al. (2021). On the Opportunities and Risks of Foundation Models. Stanford Center for Research on Foundation Models. arXiv:2108.07258. DOI: 10.48550/arXiv.2108.07258.
2. Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., et al. (2020). Language Models are Few-Shot Learners. *Advances in Neural Information Processing Systems*, 33, 1877–1901. arXiv:2005.14165.
3. Chatterjee, A., Renduchintala, H. S. V. N. K., Bhatia, S., & Chakraborty, T. (2024). POSIX: A Prompt Sensitivity Index For Large Language Models. *Findings of the Association for Computational Linguistics: EMNLP 2024*, 14550–14565. DOI: 10.18653/v1/2024.findings-emnlp.852.
4. Chen, K., Yu, X., Zhou, Y., & Wang, H. (2026). Prompt Stability Matters: Evaluating and Optimizing Auto-Generated Prompt in General-Purpose Systems. *Conference on Parsimony and Learning*, PMLR, 328, 360–374.
5. Errica, F., Sanvito, D., Siracusano, G., & Bifulco, R. (2025). What Did I Do Wrong? Quantifying LLMs' Sensitivity and Consistency to Prompt Engineering. *Proceedings of NAACL 2025*, 1543–1558. DOI: 10.18653/v1/2025.naacl-long.73.
6. Hua, A., Tang, K., Gu, C., Gu, J., Wong, E., & Qin, Y. (2025). Flaw or Artifact? Rethinking Prompt Sensitivity in Evaluating LLMs. *Proceedings of EMNLP 2025*, 19889–19899. DOI: 10.18653/v1/2025.emnlp-main.1006.
7. Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., et al. (2023). Survey of Hallucination in Natural Language Generation. *ACM Computing Surveys*, 55(12), Article 248, 1–38. DOI: 10.1145/3571730.
8. Kojima, T., Gu, S. S., Reid, M., Matsuo, Y., & Iwasawa, Y. (2022). Large Language Models are Zero-Shot Reasoners. *Advances in Neural Information Processing Systems*, 35. arXiv:2205.11916. DOI: 10.48550/arXiv.2205.11916.
9. Lin, S., Hilton, J., & Evans, O. (2022). TruthfulQA: Measuring How Models Mimic Human Falsehoods. *Proceedings of ACL 2022*, 3214–3252. DOI: 10.18653/v1/2022.acl-long.229.
10. Liu, N. F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., & Liang, P. (2024). Lost in the Middle: How Language Models Use Long Contexts. *Transactions of the Association for Computational Linguistics*, 12, 157–173. DOI: 10.1162/tacl_a_00638.
11. Liu, Y., & Chu, C. (2026). Understanding the Prompt Sensitivity. *Proceedings of ACL 2026*, 44362–44388. DOI: 10.18653/v1/2026.acl-long.2053. arXiv:2604.18389.
12. Si, C., Yang, D., & Hashimoto, T. (2024). Can LLMs Generate Novel Research Ideas? A Large-Scale Human Study with 100+ NLP Researchers. arXiv:2409.04109. DOI: 10.48550/arXiv.2409.04109.
13. Wang, L., Chen, X., Deng, X., Wen, H., You, M., Liu, W., Li, Q., et al. (2024). Prompt engineering in consistency and reliability with the evidence-based guideline for LLMs. *npj Digital Medicine*, 7, Article 41. DOI: 10.1038/s41746-024-01029-4.

14. Wang, X., Wei, J., Schuurmans, D., Le, Q. V., Chi, E. H., Narang, S., Chowdhery, A., & Zhou, D. (2023). Self-Consistency Improves Chain of Thought Reasoning in Language Models. *International Conference on Learning Representations (ICLR 2023)*. arXiv:2203.11171. DOI: 10.48550/arXiv.2203.11171.
15. Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E., Le, Q. V., & Zhou, D. (2022). Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. *Advances in Neural Information Processing Systems*, 35, 24824–24837. DOI: 10.52202/068431-1800. arXiv:2201.11903.
16. Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., et al. (2023). Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. arXiv:2306.05685. DOI: 10.48550/arXiv.2306.05685.
17. Zhuo, J., Zhang, S., Fang, X., Duan, H., Lin, D., & Chen, K. (2024). ProSA: Assessing and Understanding the Prompt Sensitivity of LLMs. *Findings of the Association for Computational Linguistics: EMNLP 2024*, 1950–1976. DOI: 10.18653/v1/2024.findings-emnlp.108.